THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Human activity can disturb Earth's: 1

(a) climate patterns **(b)** distance from Sun
(c) size of Moon **(d)** spin direction only

2. Which activity shows scientific attitude? 1

(a) Copying answers **(b)** Checking claims with observation
(c) Ignoring evidence **(d)** Believing everything

3. Which pair is correctly matched? 1

(a) Yeast – nitrogen fixation **(b)** Rhizobium – bread
(c) Amoeba – biogas **(d)** Lactobacillus – curd

4. Vaccines are: 1

(a) poisonous **(b)** antibiotics **(c)** curative **(d)** preventive

5. Staying at home and resting when unwell helps to: 1

(a) recover and limit spread **(b)** spread disease
(c) lose immunity **(d)** avoid medicine forever

6. The heat produced in a wire increases when: 1

(a) the circuit is broken **(b)** the wire is cooled
(c) the current increases **(d)** the current decreases

7. Metal rods in a cell are called: 1

(a) filaments **(b)** coils **(c)** fuses **(d)** electrodes

8. For a force to act, there must be at least: 1

(a) no objects **(b)** two interacting objects
(c) three objects **(d)** one object

9. Bordoila is a local storm of: 1

(a) Sikkim **(b)** Gujarat **(c)** Goa **(d)** Assam

10. A nail is driven easily by its pointed end because: 1

(a) the force is zero **(b)** the pressure is zero
(c) the area is large **(d)** the area is small so pressure is large

11. Particles of matter are held together by: 1

(a) interparticle forces of attraction **(b)** magnetic poles only
(c) gravity of Moon **(d)** friction only

12. Which is a pure substance? 1

(a) Soil **(b)** Sea water **(c)** Milk **(d)** Distilled water

13. Which of these is an element? 1

(a) Rust **(b)** Oxygen **(c)** Water **(d)** Salt

14. When ice forms, water particles: 1

(a) turn to gas **(b)** arrange to take up more space
(c) come closer **(d)** disappear

15. A convex lens makes light rays: 1

(a) converge **(b)** stay parallel **(c)** diverge **(d)** stop

16. Sun rises exactly in the east on: 1

(a) all days **(b)** equinoxes **(c)** new moon days **(d)** solstices

17. Ecosystem services include: 1

(a) clean air, water, food and climate regulation
(b) factories and cars
(c) highways
(d) mines

18. A pond is an example of an: 1

(a) grassland **(b)** aquatic ecosystem
(c) desert ecosystem **(d)** terrestrial ecosystem

19. Earth's nearly circular orbit ensures that: 1

(a) there is no day
(b) temperature is not too extreme during the year
(c) Earth is always hot
(d) no seasons exist

20. Local communities can help sustain Earth by: 1

(a) burning forests **(b)** wasting water
(c) using natural resources sustainably **(d)** polluting rivers

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What did Ananda Mohan Chakrabarty develop? 2

22. What is variolation? 2

23. Mention two ways Earth's magnetism is useful. 2

24. State the key principle for the formation of wind. 2

25. What is Mishraloha? 2

26. Find the volume of 500 g of a substance with density 2 g/cm³. 2

27. What are invasive species? 2

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28. State two ways to reduce pollution. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Why does the book end with Earth and climate? 3

30. Give three examples of muscular force in daily life and in the body. 3

31. Why does a gas exert pressure on the walls of a container? 3

32. Why do we see lateral inversion in all three types of mirrors? Give an example. 3

33. State three uses of calendars in society. 3

34. Why do small changes in temperature matter for life on Earth? 3

35. What is the role of the carbon rod and zinc container in a dry cell? 3

OR

Explain how the poles of an electromagnet can be found using a compass. 3

36. State three ways in which students can help conserve ecosystems. 3

OR

Explain why the number of organisms decreases at higher trophic levels. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. Prepare a table for any four communicable diseases with agent, symptom and preventive measure. 5

OR

Describe how antibiotic resistance occurs and how it spreads. Suggest four ways to control it. 5

38. Explain how a lightning conductor works with a diagram. 5

OR

Describe the process of lightning formation. State four safety rules. 5

39. A solid block has mass 480 g and volume 160 cm^3. 5

(i) Find density.

(ii) Will it float in water?

(iii) Find mass of 50 cm^3 of it.

OR

Explain solute, solvent, solution, saturated, unsaturated and solubility with examples. 5

40. Explain the problems caused by pesticides and monoculture and suggest sustainable alternatives. 5

OR

Explain trophic levels and the pyramid. Describe energy flow from producers to top consumers. 5

END OF QUESTION PAPER Total: 80 Marks